HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python**-based AI agents and agentic workflows for automating data processing and business intelligence reporting, handling approximately **5,000 daily events**.
- Built and integrated **LLM-powered applications** using **LangChain** and **OpenAI APIs**, optimizing prompts and context management to improve response accuracy by an average of **15%**.
- Engineered **RAG solutions** incorporating **vector databases** and custom embeddings to combine structured and unstructured enterprise data, enhancing data retrieval relevance by **22%**.
- Implemented secure **API integrations** using the Model Context Protocol (MCP) to connect AI systems with internal databases and third-party services, ensuring data consistency across systems.
- Managed the full lifecycle of AI features from prototype to production, including implementing **automated testing**, robust **monitoring** with Splunk, and safe rollback strategies.
- Applied **MLOps/LLMOps best practices**, including version control with Git, CI/CD pipelines with Jenkins, and containerisation using **Docker** for reliable deployment.
- Designed and implemented **audit trails** and access controls around AI agents handling sensitive data, adhering to security, privacy, and responsible AI standards.

SKILLS

AI/ML Frameworks – LangChain, LlamaIndex, OpenAI API, Azure OpenAI, Anthropic Claude, scikit-learn

Languages – Python, Java, SQL, R

Cloud Platforms – Azure AI, AWS (EC2, S3, Lambda, CloudWatch)

Databases – Vector Databases, PostgreSQL, MySQL, Oracle

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Splunk, Dynatrace, FastAPI, Spring Boot, Power BI, Tableau

Concepts – LLM, RAG, Embeddings, Prompt Engineering, MLOps, LLMOps, CI/CD, API Integration, Data Pipelines, Data Modelling

PROJECTS

AI Agent for Automated Business Reporting

- Developed an AI agent using **Python** and **LangChain** to automate the generation of weekly business intelligence reports, reducing manual effort by approximately **30%**.
- Integrated the agent with **PostgreSQL** and internal APIs to fetch and process structured data, presenting insights through a dashboard interface.
- Implemented logging and error handling mechanisms to ensure reliability and provide clear audit trails for automated report generation processes.

RAG-based Knowledge Retrieval System

- Built a retrieval-augmented generation (RAG) system using **Python**, a **vector database** (Pinecone), and custom embeddings to enable efficient querying of enterprise documentation.
- Engineered data pipelines to ingest and chunk large volumes of unstructured text data, optimizing for search relevance and retrieval speed.
- Developed a FastAPI service to expose the RAG system's capabilities via a REST API, allowing other internal applications to query the knowledge base.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)